💨 Compression: Reading

Small Engines Overhaul · Unit 4 · Level 3 (closer to the class work) · goes with the Compression Handout and article

Name: _____

Compression: The Squeeze That Makes Power

Compression is the squeezing of the air-fuel mixture during the second stroke. When the mixture is squeezed into a small space and then ignited, it burns faster and builds more pressure, pushing the piston down with more force. Good compression is needed for easy starting and full power.

Several parts seal the combustion chamber. Piston rings seal the piston to the cylinder wall. The intake and exhaust valves must close tightly against their seats. The head gasket seals the cylinder head to the block, and the spark plug seals the plug hole.

When any seal fails, pressure leaks out. Common causes are worn or stuck rings, a scored cylinder, burned or carbon-covered valves, incorrect valve clearance and a blown head gasket.

A compression test checks the seal. With the spark plug removed, a compression gauge is screwed into the plug hole and the engine is turned over several times. The reading in PSI is compared to the manufacturer's specification. Many small engines have a compression release that lowers the reading, so always use the manual's spec.

Check your reading

1. Why does compression increase power?

2. Name three parts that seal the combustion chamber.

3. How is a compression test done?